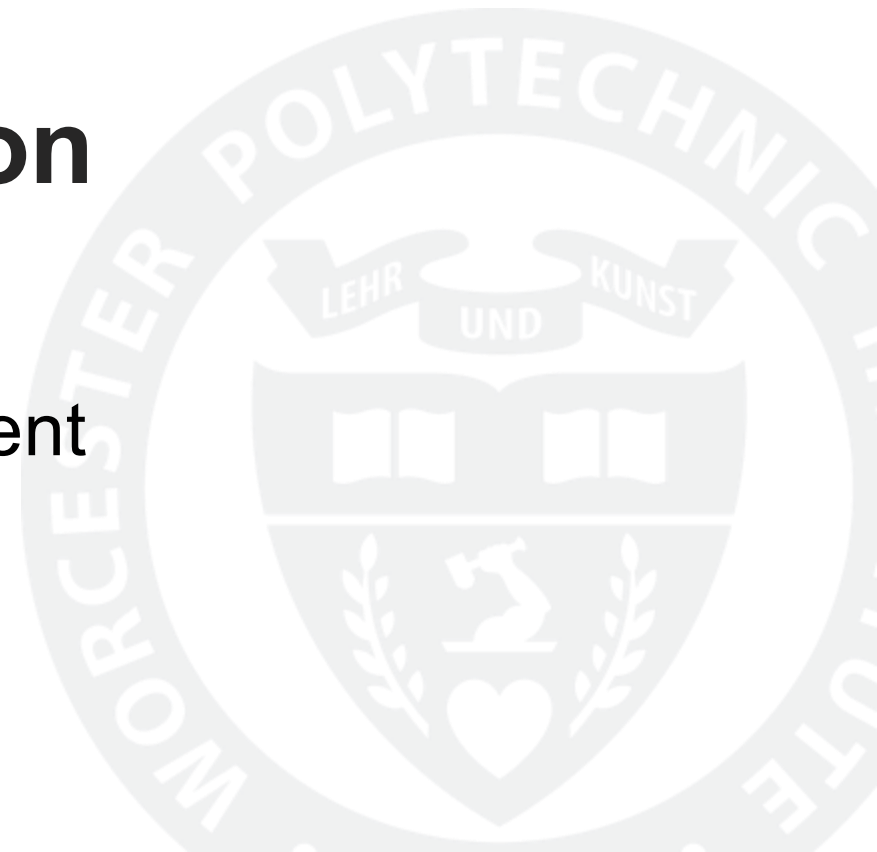




WPI

Cross-Vehicle Online Adaptation for Off-road Traversability

Motivation, Literature Review, Problem Statement



Overview

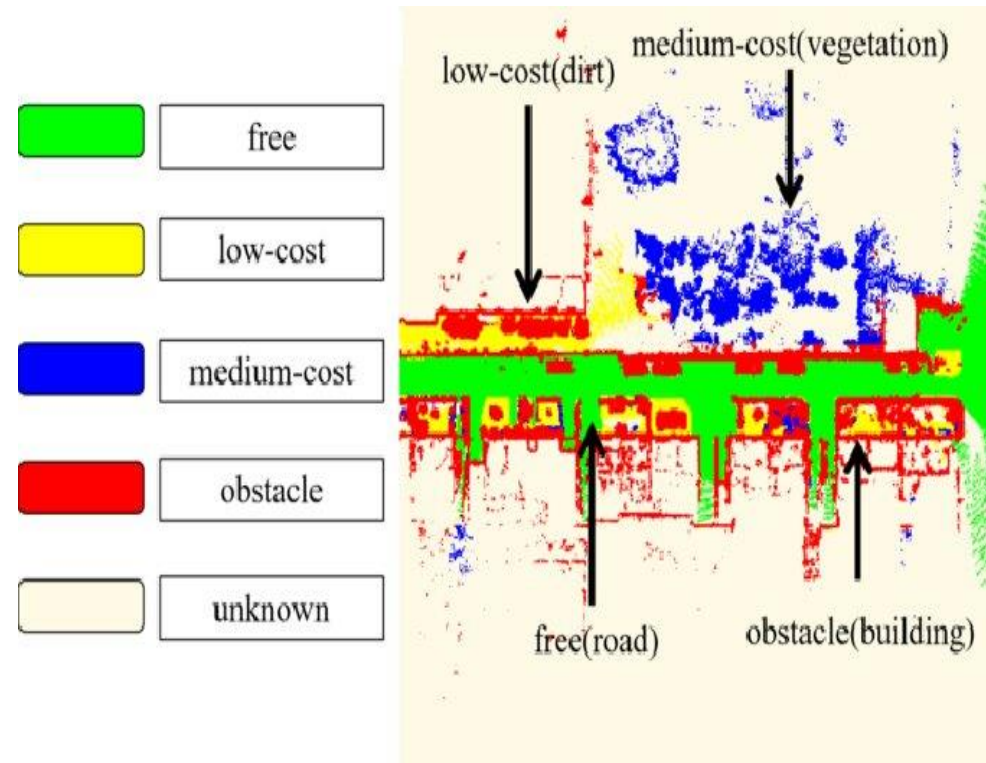
- Motivation
- Literature Review
- Problem Statement

Overview

- Motivation
- Literature Review
- Problem Statement

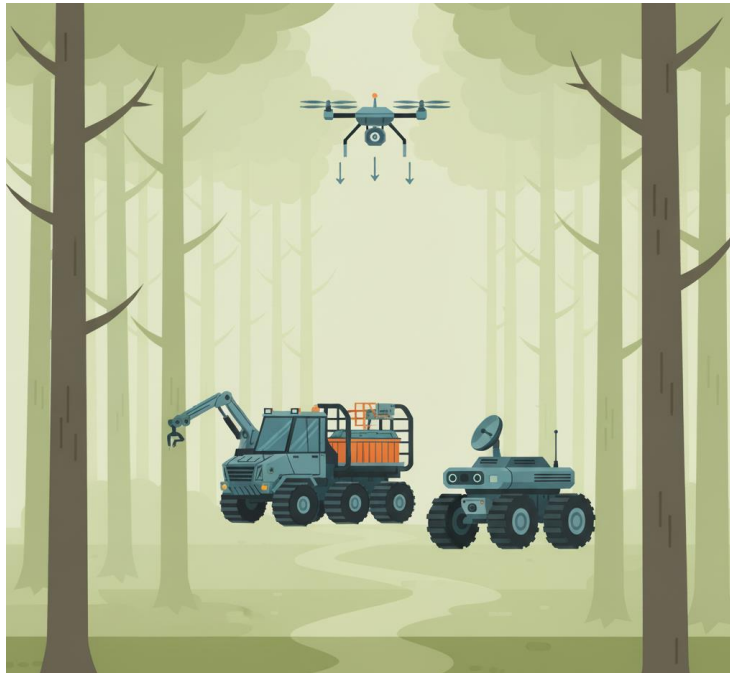
Motivation for Offroad Traversability Mapping

- Applications
 - Search & Rescue
 - Planetary Exploration
 - Forestry
- Challenges
 - Unstructured terrain
 - Uneven terrain
 - Unseen terrain
 - Small datasets
 - Robot specific



Motivation: Heterogeneous Robot Exploration

- Different robots = different traversability
- How can we transfer traversability maps between vehicles?
- How can we combine drone scan and ground vehicle data?



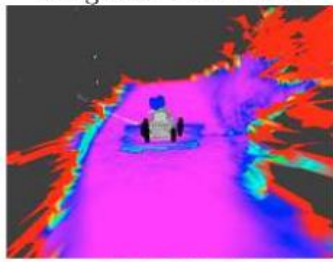
Overview

- Motivation
- Literature Review
- Problem Statement

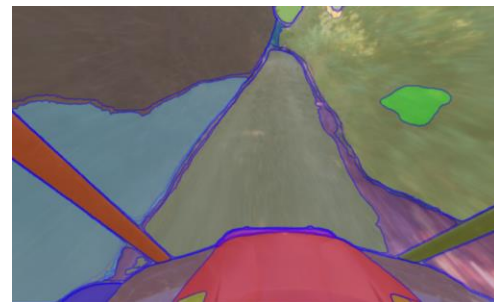
Literature Review – Overall Methods



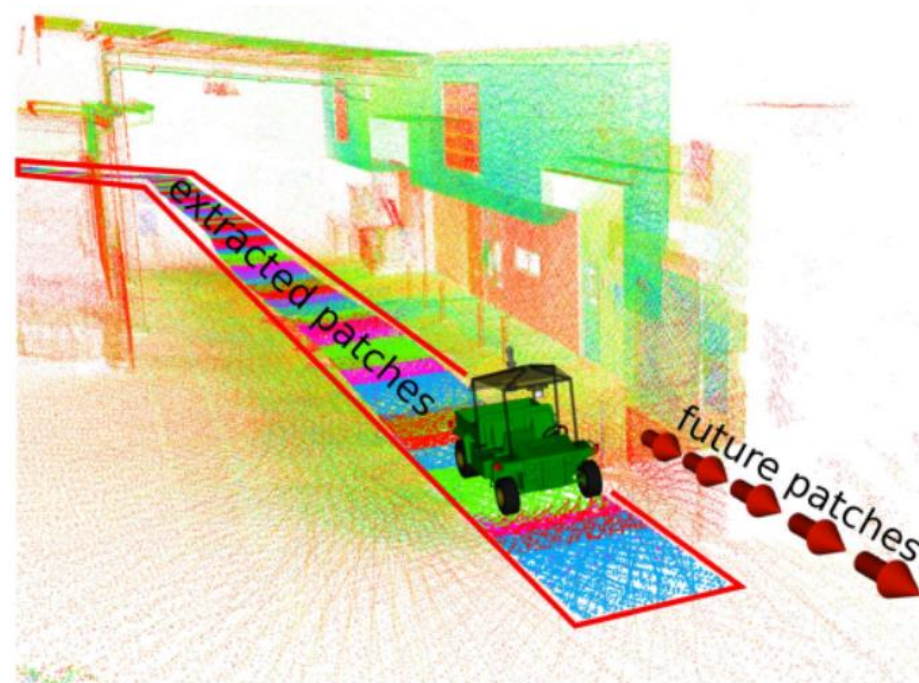
Roughness Estimation



Geometric Methods



Segmentation Methods



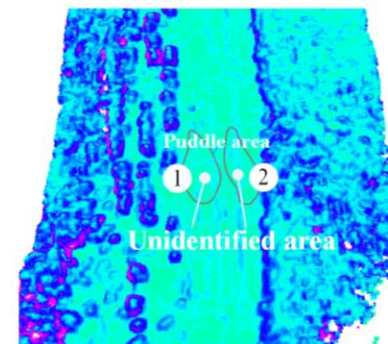
Proprioceptive Self Supervision

Geometric Methods

- Captures physical geometry
- Requires hand-made features
- Cannot detect puddles

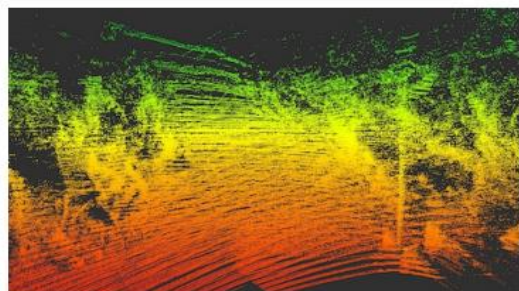


Camera

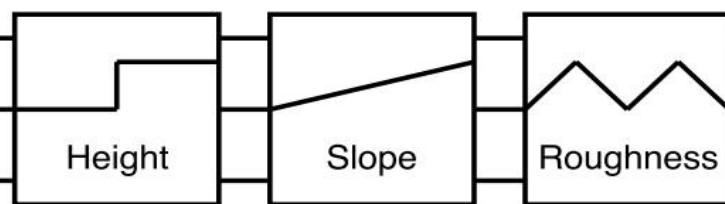


Geometric Map

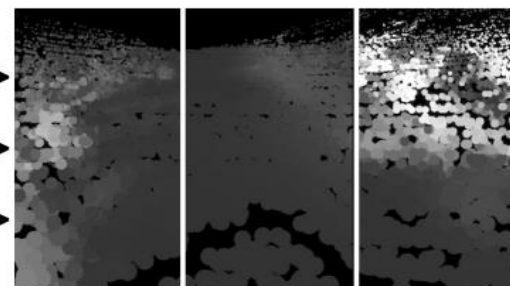
[3]



Registered point cloud



Feature extraction



Geometric features

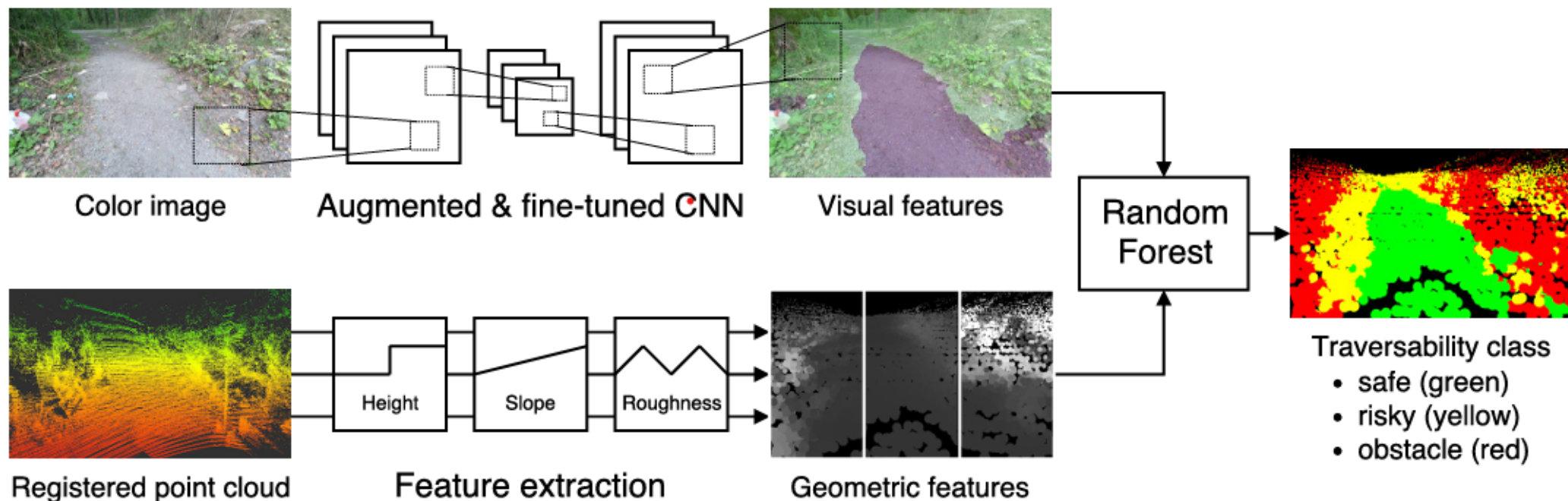
[2]

[1] "3d" navigation mesh generation for path planning in uneven terrain," (IFAC 2016)

[2] Geometric and Visual Terrain Classification for Autonomous Mobile Navigation (IROS 2017)

[3] Autonomous Vehicles Traversability Mapping Fusing Semantic-Geometric in Off-Road Navigation (Drones 2024)

Geometric Features + Semantic Segmentation



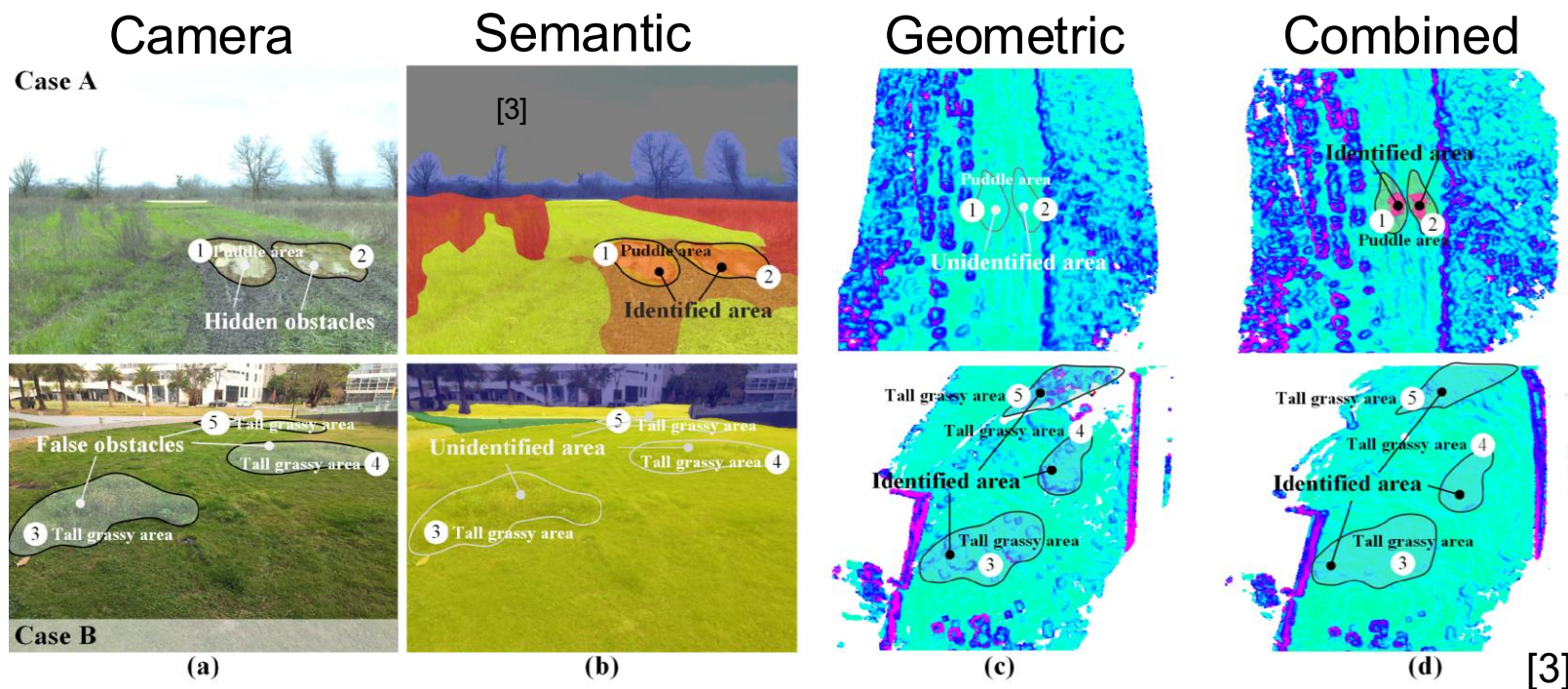
[2]

[2] Geometric and Visual Terrain Classification for Autonomous Mobile Navigation (IROS 2017)

[3] Autonomous Vehicles Traversability Mapping Fusing Semantic-Geometric in Off-Road Navigation (Drones 2024)

Geometric Methods + Semantic Segmentation

- Can detect puddles
- Can filter out long grass
- But requires traversability knowledge



[2] Geometric and Visual Terrain Classification for Autonomous Mobile Navigation (IROS 2017)

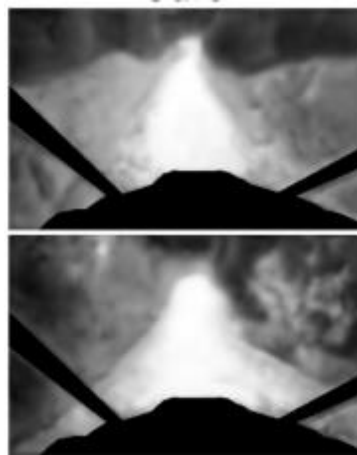
[3] Autonomous Vehicles Traversability Mapping Fusing Semantic-Geometric in Off-Road Navigation (Drones 2024)

Segmentation Methods – Self Supervised

- Inflate traversable region with segmentation
- Measure similarity to "traversable vector"



Camera



Traversability



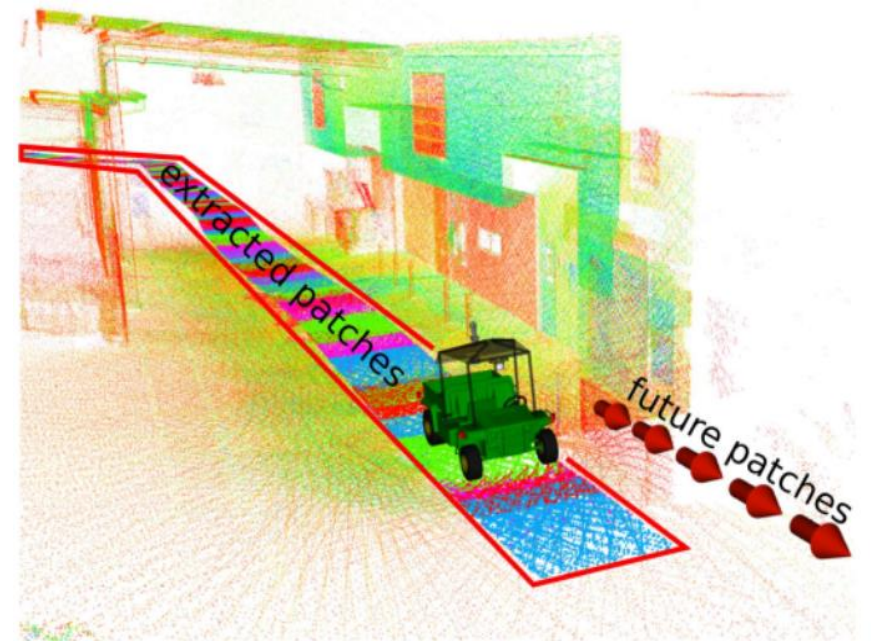
Projected Path



Mark full segment as Traversable

Proprioceptive Self Supervision

- Record future patches
 - Camera
 - LIDAR
- Drive over a patch and record traversability
 - Velocity
 - IMU
 - Wheel odometry
- Train a model
 - Camera + LIDAR patch → traversability

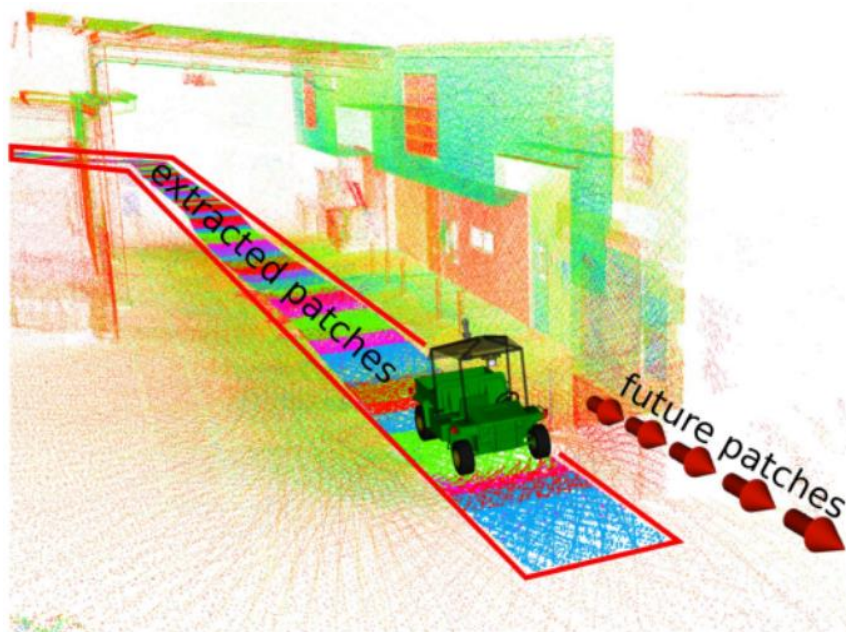


[8]

- [6] From Simulation to Field: Learning Terrain Traversability for Real-World Deployment (2025)
- [7] How Does It Feel? Self-Supervised Learning for Off-Road Vehicle Traversability (ICRA 2023)
- [8] How Rough Is the Path? Terrain Traversability Estimation for Local and Global Path Planning (IEEE-TITS 2022)
- [9] TerraPN: Unstructured Terrain Navigation using Online Self-Supervised Learning (IROS 2022)
- [10] Learning multiobjective rough terrain traversability (Journal of Terramechanics 2022)
- [11] Learning traversability models for autonomous mobile vehicles (Autonomous Robots 2007)

Proposed Approach

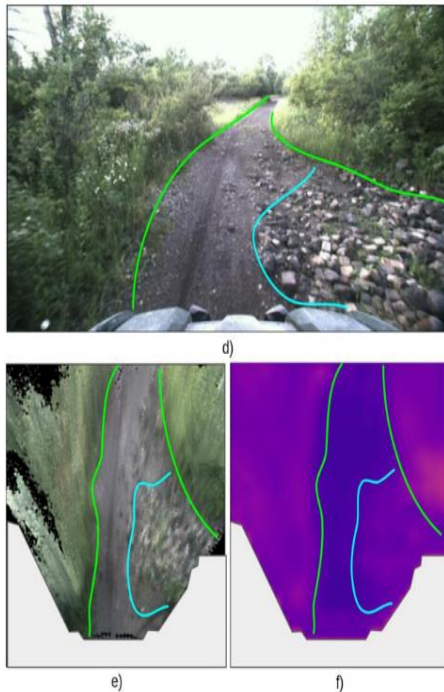
- Use proprioceptive self supervision
- Inflate labels with segmentation



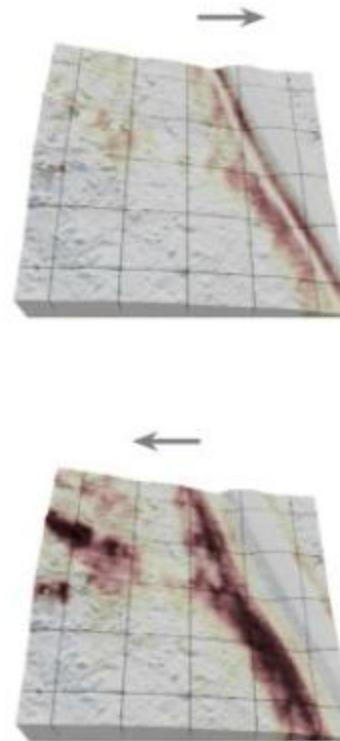
Traversability Modeling



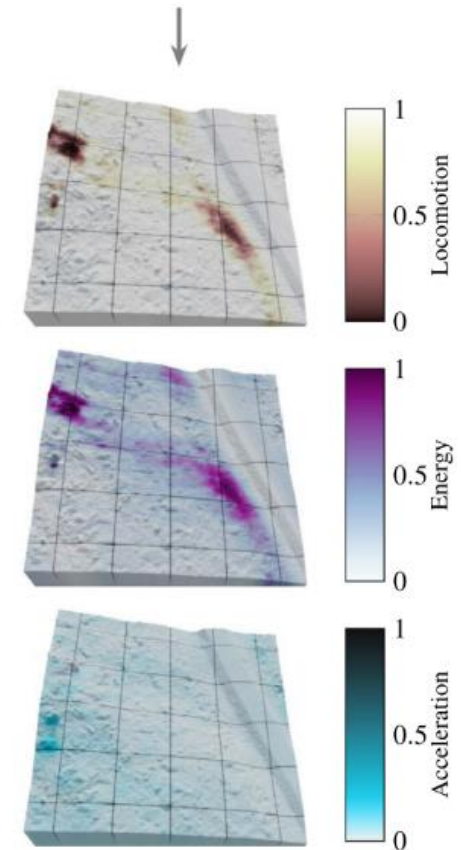
Binary/Ternary



Scalar



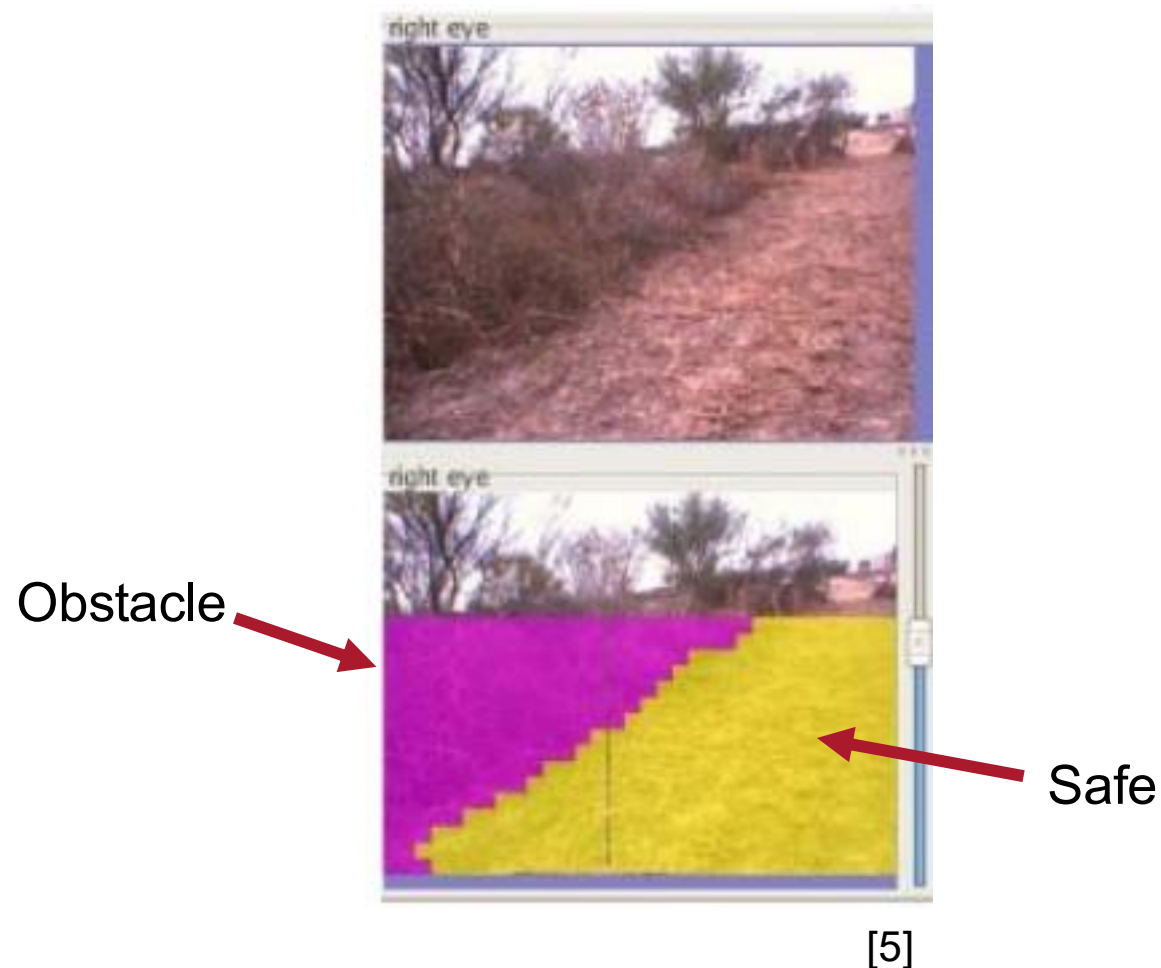
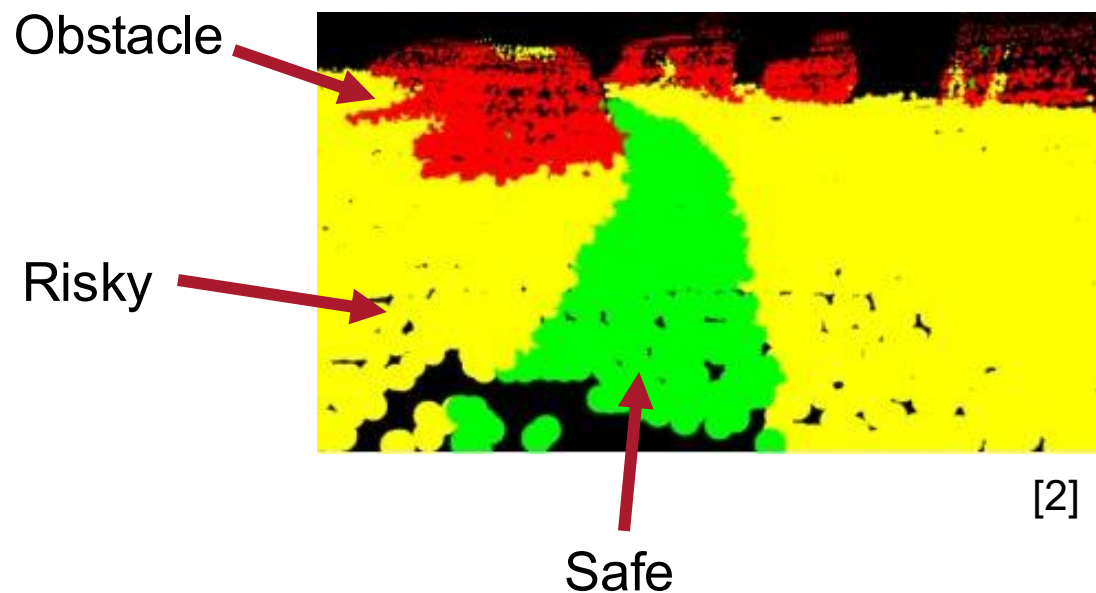
Directional



Different Metrics

Binary/Ternary Traversability

- Good for defined paths

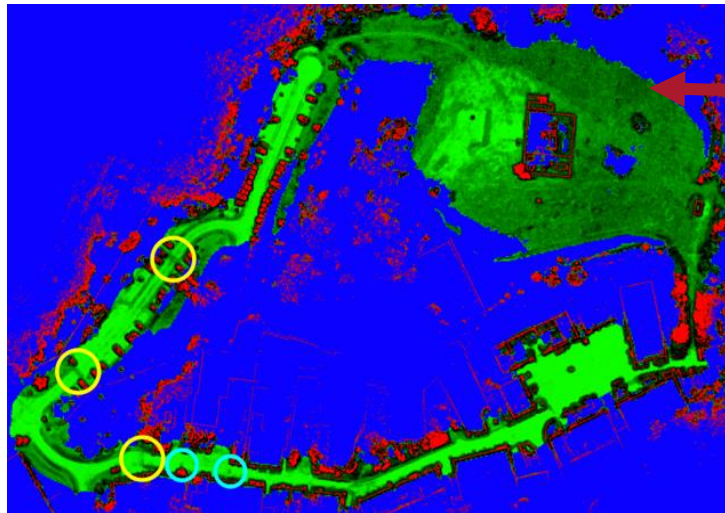


[2] Geometric and Visual Terrain Classification for Autonomous Mobile Navigation (IROS 2017)

[5] Learning traversability models for autonomous mobile vehicles (Autonomous Robots 2007)

Scalar Traversability

- More expressive for offroad
- [4][7][9][11] create local maps
- [8] creates a global map

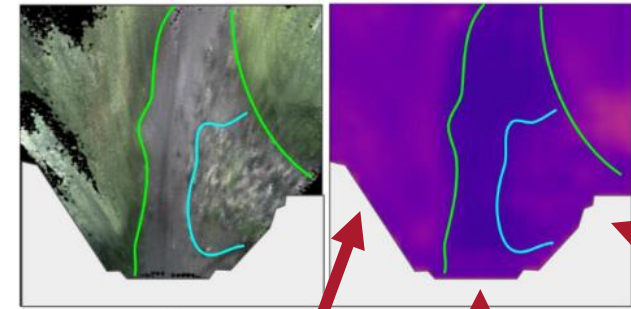


[8]

Hilly terrain



d)



Bushes

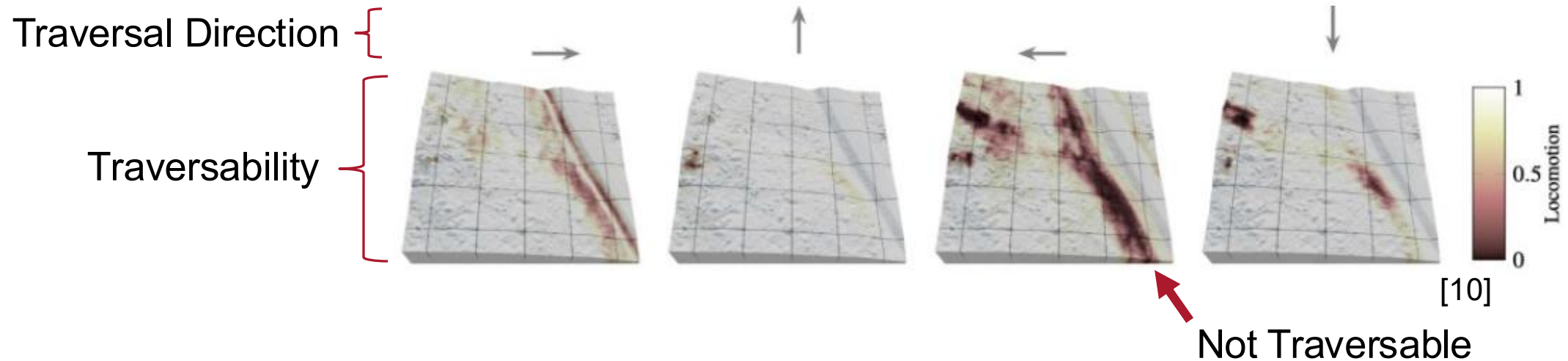
Path

Tree
Gravel

[7]

- [4] V-STRONG: Visual Self-Supervised Traversability Learning for Off-road Navigation (ICRA 2024)
[7] How Does It Feel? Self-Supervised Learning for Off-Road Vehicle Traversability (ICRA 2023)
[8] How Rough Is the Path? Terrain Traversability Estimation for Local and Global Path Planning (IEEE-TITS 2022)
[9] TerraPN: Unstructured Terrain Navigation using Online Self-Supervised Learning (IROS 2022)
[11] METAverse: Meta-Learning Traversability Cost Map for Off-Road Navigation IROS 2024

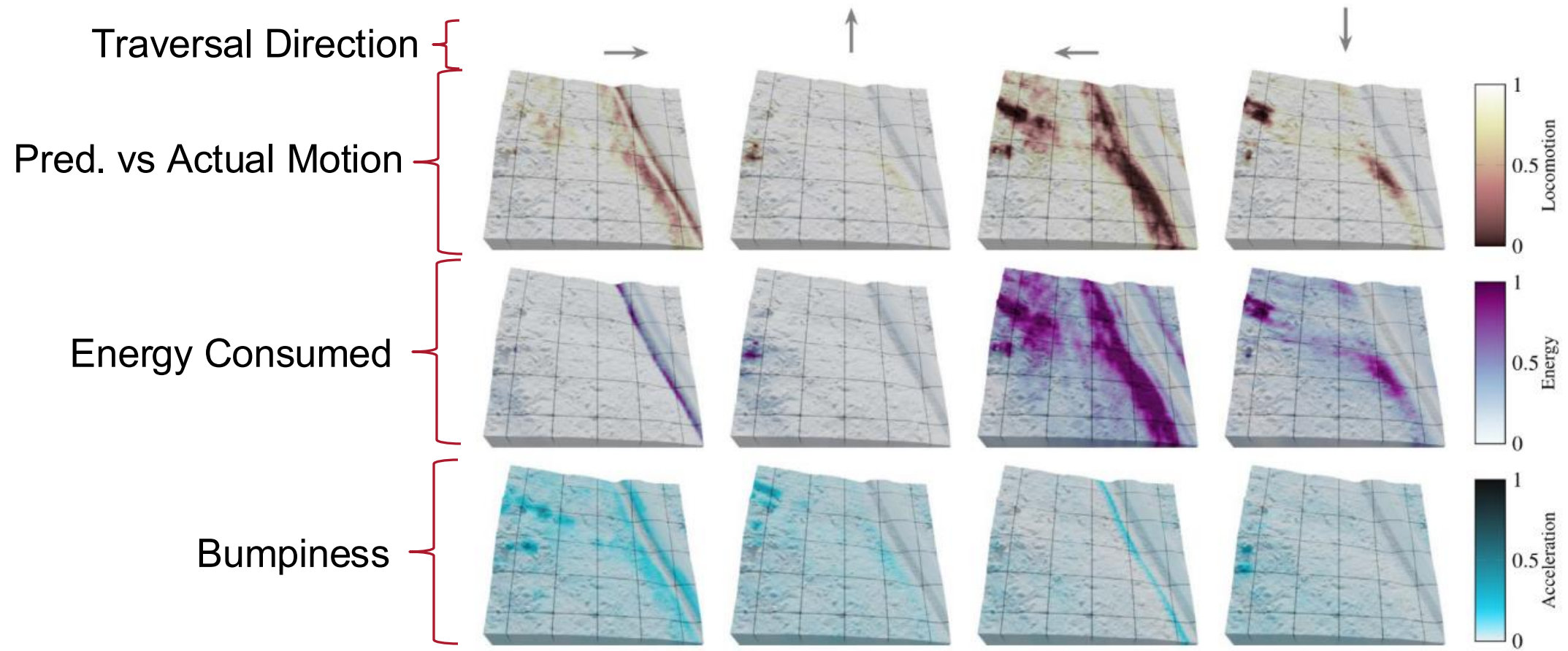
Directional Traversability



[6] From Simulation to Field: Learning Terrain Traversability for Real-World Deployment (2025)

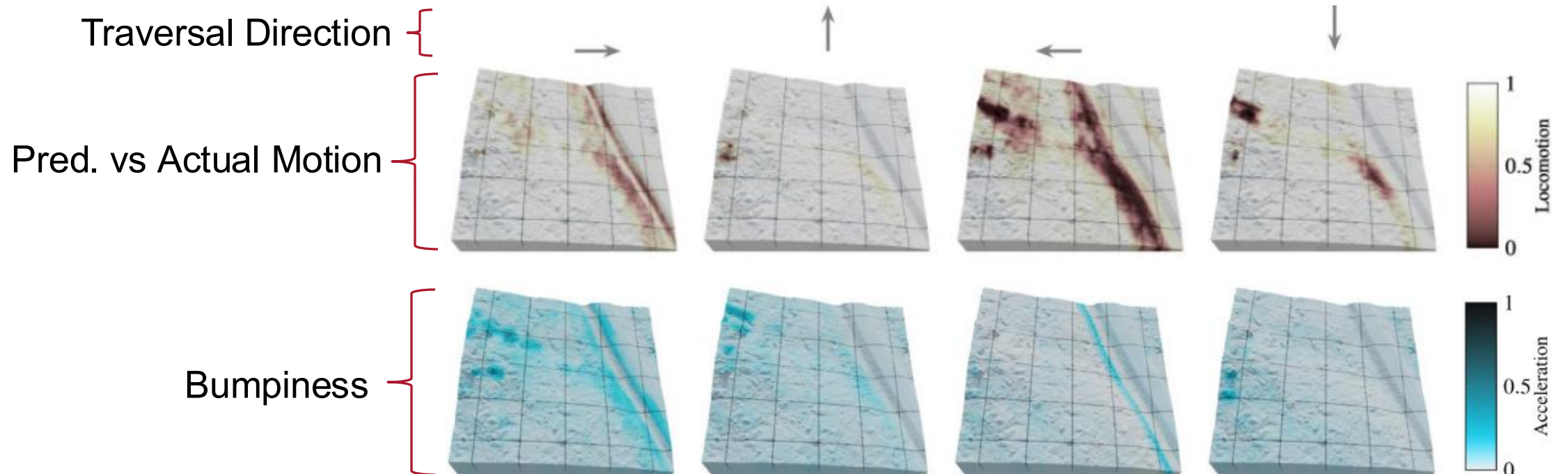
[10] Learning multiobjective rough terrain traversability (Journal of Terramechanics 2022)

Traversability Metrics



Proposed Approach

- Global Mapping
- Directional Traversability
- Combine Predicted vs Actual Motion and Bumpiness



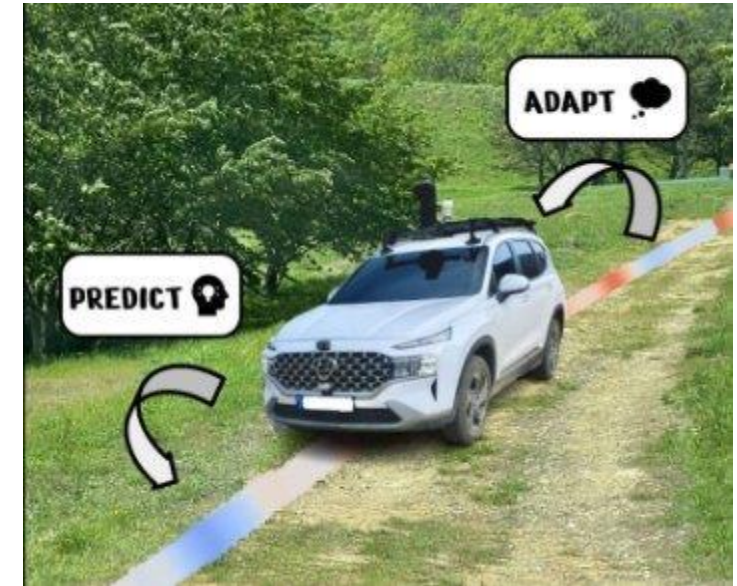
Literature Review – Data Collection



Simulation



Offline
Real Life



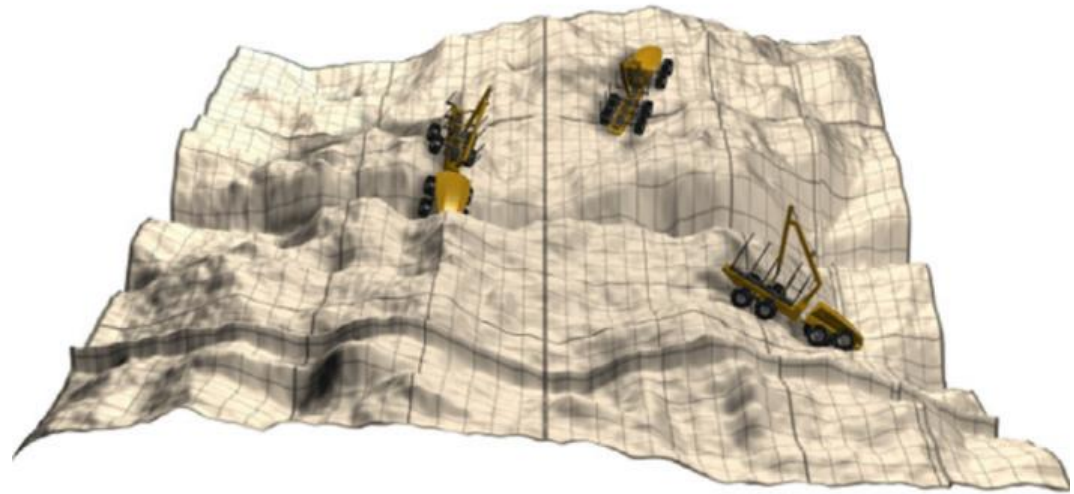
Online
Real Life

Data Collection in Simulation

- Can test dangerous situations
- Can gather lots of data
- Sim2Real gap



[6]



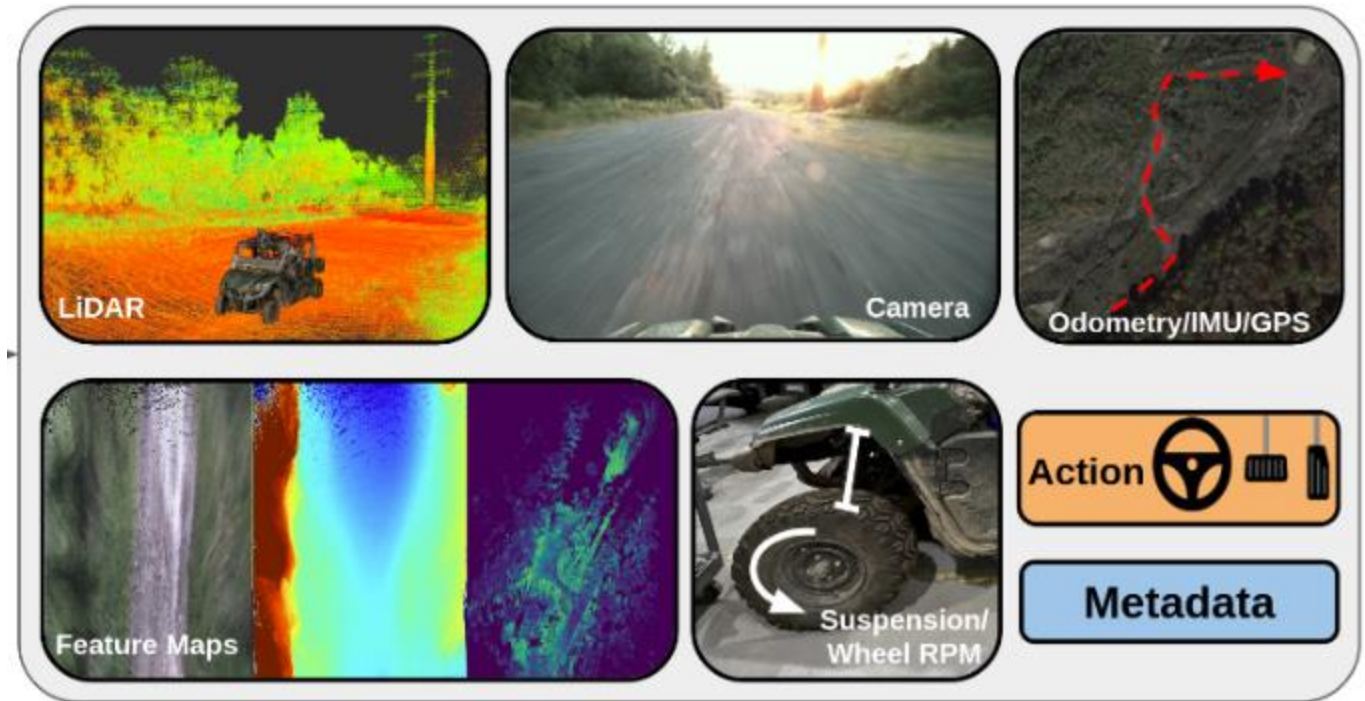
[10]

[6] From Simulation to Field: Learning Terrain Traversability for Real-World Deployment (2025)

[10] Learning multiobjective rough terrain traversability (Journal of Terramechanics 2022)

Training on Offline Real-Life Data

- Real-World physics
- Limited to vehicle(s) in dataset
- No dangerous situations



TartanDrive 2.0 [12]

[2] Geometric and Visual Terrain Classification for Autonomous Mobile Navigation (IROS 2017)

[4] V-STRONG: Visual Self-Supervised Traversability Learning for Off-road Navigation (ICRA 2024)

[7] How Does It Feel? Self-Supervised Learning for Off-Road Vehicle Traversability (ICRA 2023)

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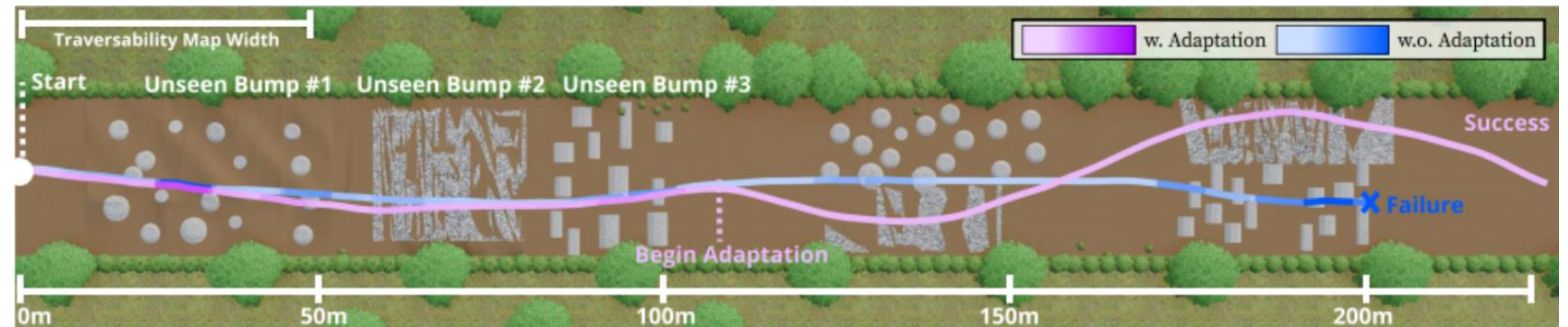
[12] TartanDrive 2.0: More Modalities and Better Infrastructure to Further Self-Supervised Learning Research in Off-Road Driving Tasks

Online Real World Collection

- Combat challenge of new environments
- Need for safety



[9]



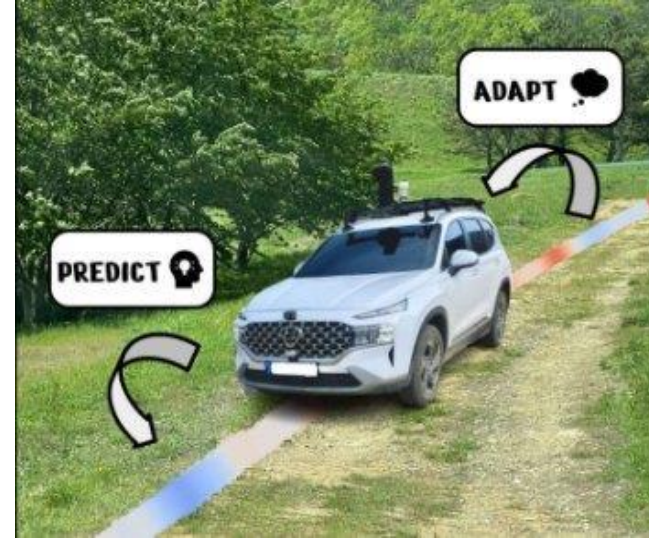
[11]

[9] TerraPN: Unstructured Terrain Navigation using Online Self-Supervised Learning (IROS 2022)

[11] METAVerse: Meta-Learning Traversability Cost Map for Off-Road Navigation (IROS 2024)

Proposed Approach – Data Collection

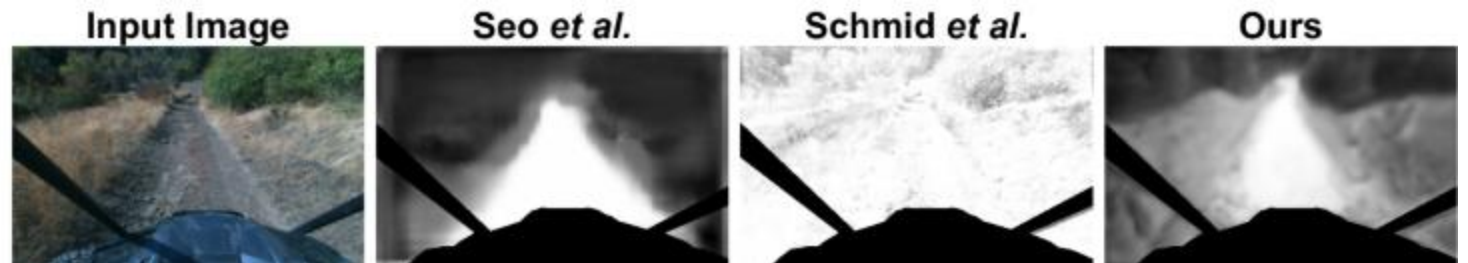
- Gather simulation data offline for two different robots
- Gather online data for robot 1
 - Transfer learning to robot 2



Evaluation - Subjective



Visual Inspection of trajectory [8]



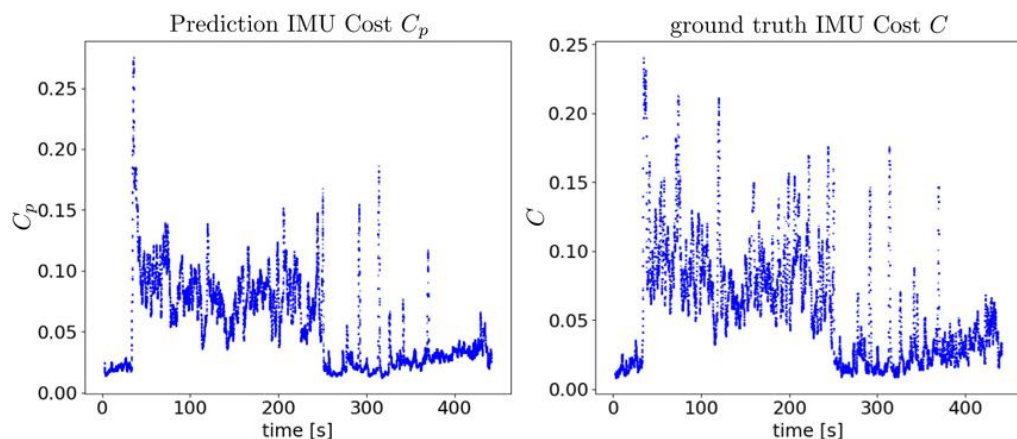
Visual inspection of traversability map [4]

[4] V-STRONG: Visual Self-Supervised Traversability Learning for Off-road Navigation (ICRA 2024)

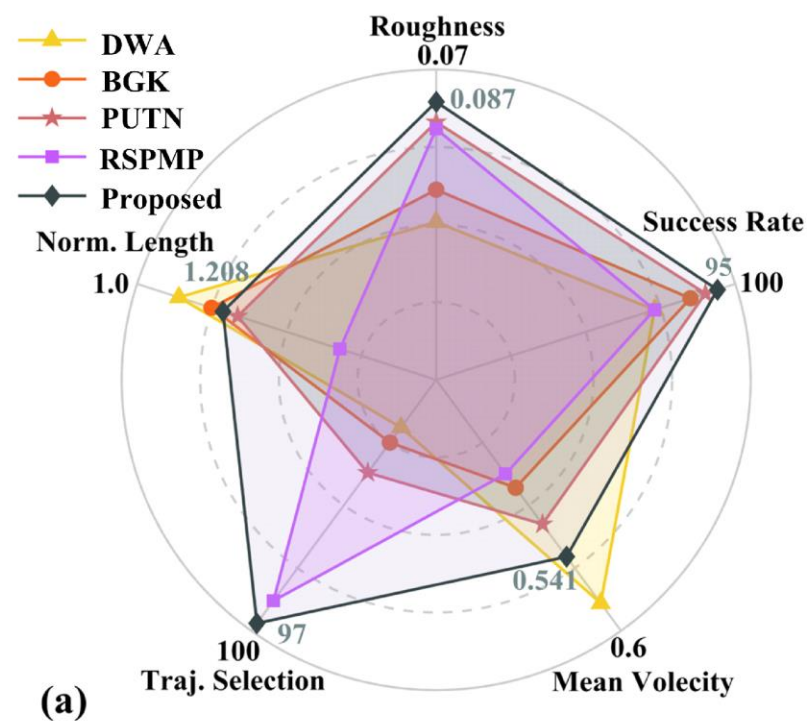
[8] How Rough Is the Path? Terrain Traversability Estimation for Local and Global Path Planning (IEEE-TITS 2022)

Evaluation - Objective

- Plan a path from start to goal
 - Measure path metrics
 - Compare against baselines
- Model Error



[8]



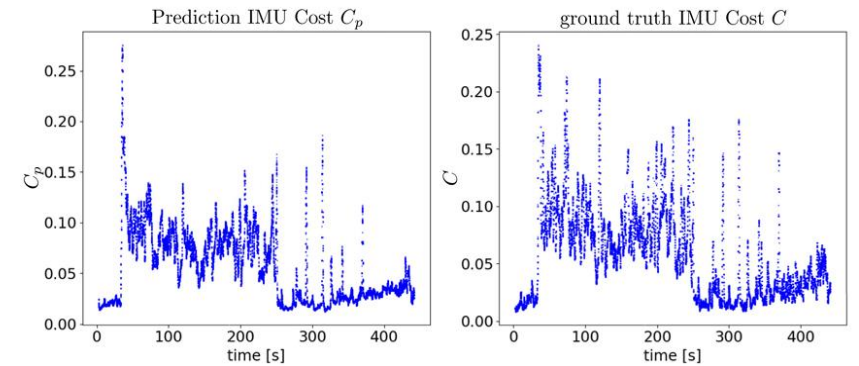
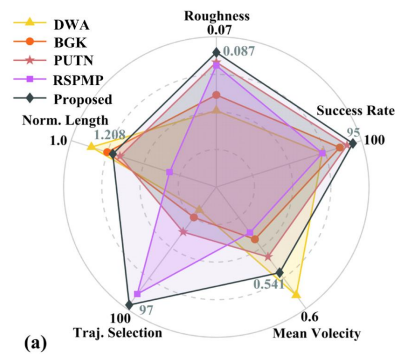
[3]

[3] Autonomous Vehicles Traversability Mapping Fusing Semantic-Geometric in Off-Road Navigation (Drones 2024)

[8] How Rough Is the Path? Terrain Traversability Estimation for Local and Global Path Planning (IEEE-TITS 2022)

Proposed Approach - Evaluation

- Visual Inspection of Traversability Map
- Objective path measures
- Model error



Literature Review – Overview

Paper	Method	Traversability Model	Data Collection	Evaluation
[1]	Geometric	Scalar, local	None	Visual Inspection
[2]	Geometric, Segmentation	Ternary, local	Offline, Real Life	Model Error
[3]	Geometric, Segmentation	Scalar, local	None	Obj. Traj. Measures
[4]	Segmentation Self Sup.	Scalar, local	Offline, Real Life	Model Error
[5]	Segmentation	Binary	None	Visual Inspection
[6]	Proprioceptive Self Sup.	Directional, global	Simulation	Model Error
[7]	Proprioceptive Self Sup.	Scalar, local	Offline, real life	Obj. Traj. Measures
[8]	Proprioceptive Self Sup.	Scalar, global	Offline, real life	Obj. Traj. Measures
[9]	Proprioceptive Self Sup.	Scalar, local	Online, real life	Obj. Traj. Measures
[10]	Proprioceptive Self Sup.	Directional, global	Simulation	Model Error
[11]	Proprioceptive Self Sup.	Scalar, local	Online, real life	Obj. Traj. Measures
Proposed	Proprio. + Seg. Self Sup.	Directional, global	Sim. Online	Obj. Traj. Measures

Literature Review – Overview

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[2]	Geometric, Segmentation	Ternary, local	Offline, Real Life	Model Error
[3]	Geometric, Segmentation	Scalar, local	None	Obj. Traj. Measures
[4]	Segmentation	Scalar, local	Offline, Real Life	Model Error
[5]	Segmentation	Binary	None	Visual Inspection

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[7]	Proprioceptive Self Sup.	Scalar, local	Offline, real life	Obj. Traj. Measures
[8]	Proprioceptive Self Sup.	Scalar, global	Offline, real life	Obj. Traj. Measures
[9]	Proprioceptive Self Sup.	Scalar, local	Online, real life	Obj. Traj. Measures
[10]	Proprioceptive Self Sup.	Directional, global	Simulation	Model Error
[11]	Proprioceptive Self Sup.	Scalar, local	Online, real life	Obj. Traj. Measures

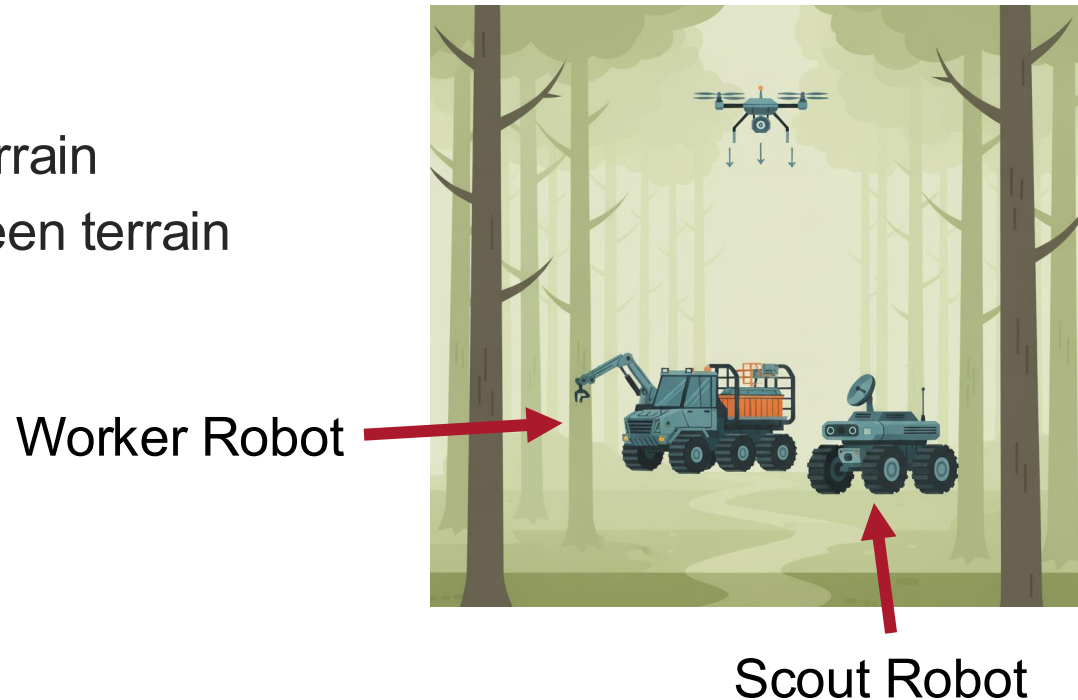
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Overview

- Motivation
- Literature Review
- Problem Statement

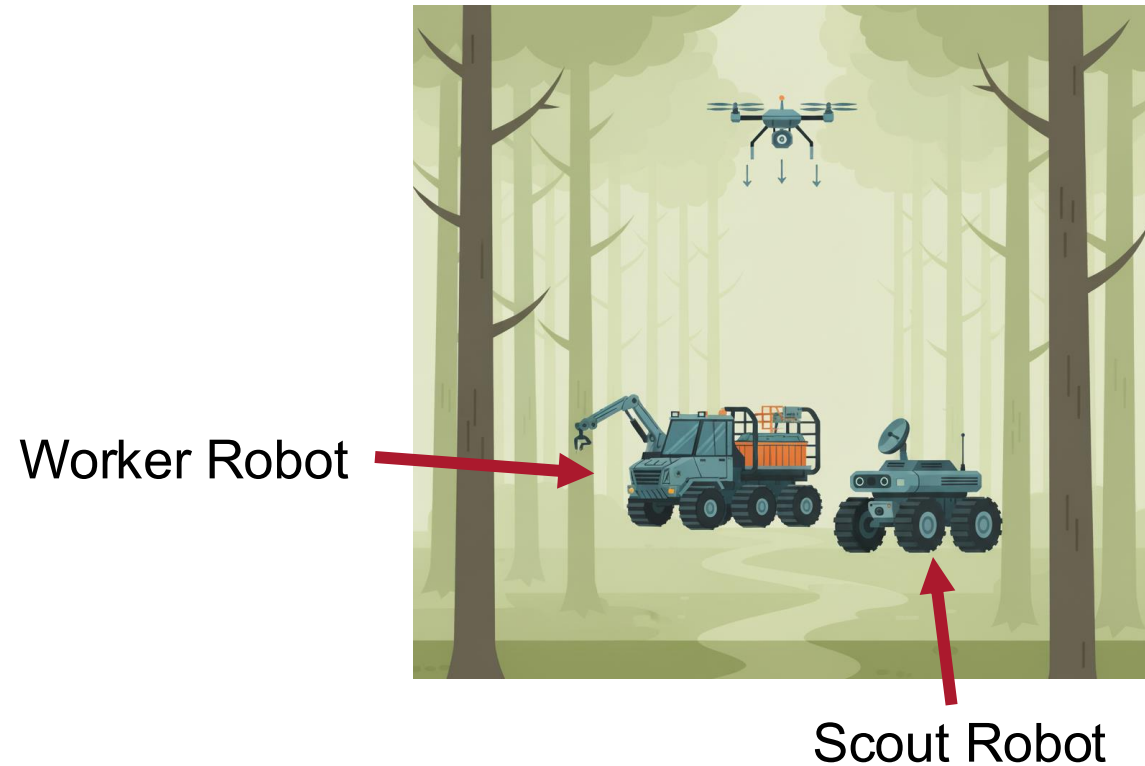
Problem Statement – Assumptions

- We have two robots with different kinodynamics
 - Weight, wheelbase, thrust, suspension
- The robots can each traverse terrain that the other cannot
- The localization problem is solved
- The evaluation terrain is
 - *Physically similar* to seen terrain
 - But *visually different* from seen terrain



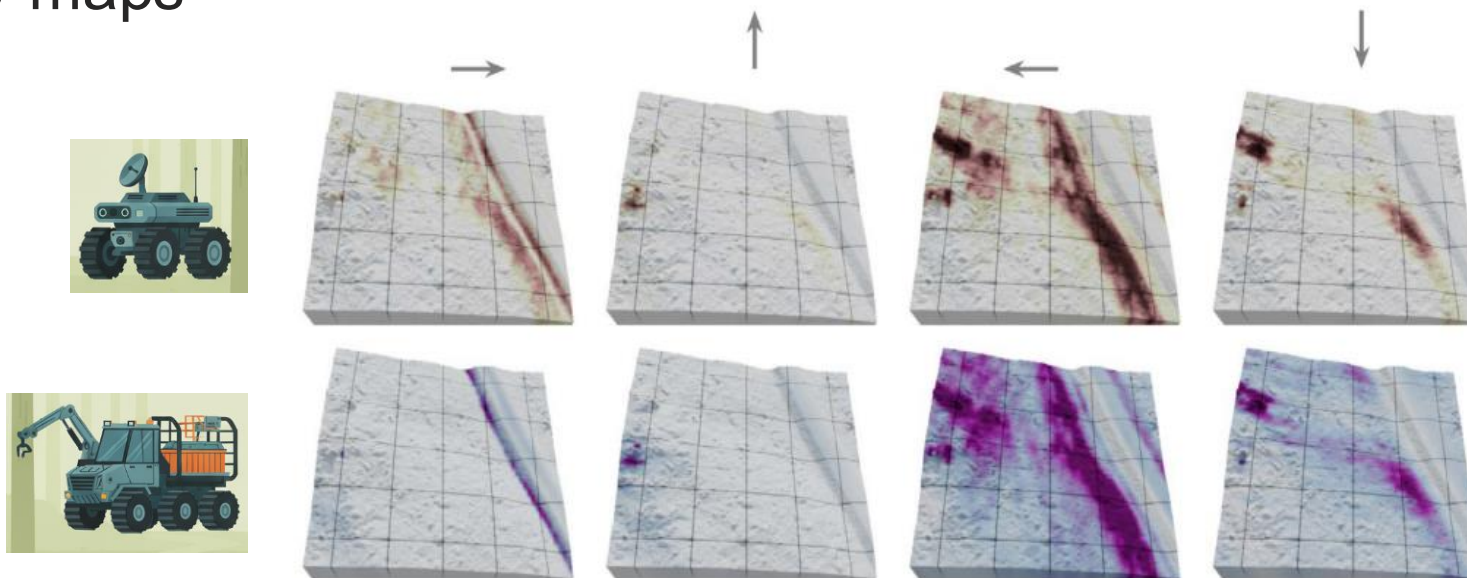
Problem Statement – Inputs

- Drone scans
 - Partial maps
 - Low resolution
- Proprioceptive
 - IMU, wheel odometry
- Exteroceptive
 - RGB-D Camera, LIDAR
- Offline Only
 - Worker Robot proprioceptive data
- Online and Offline
 - Exteroceptive data
 - Scout Robot proprioceptive data



Problem Statement – Output Traversability Maps

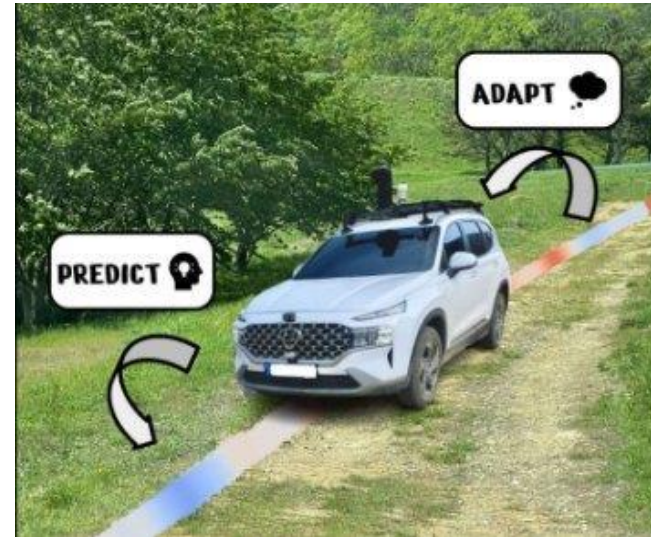
- Output two traversability maps
 - Scout Robot Map
 - Worker Robot Map
- Format
 - SE(2)
 - Confidence



Confidence

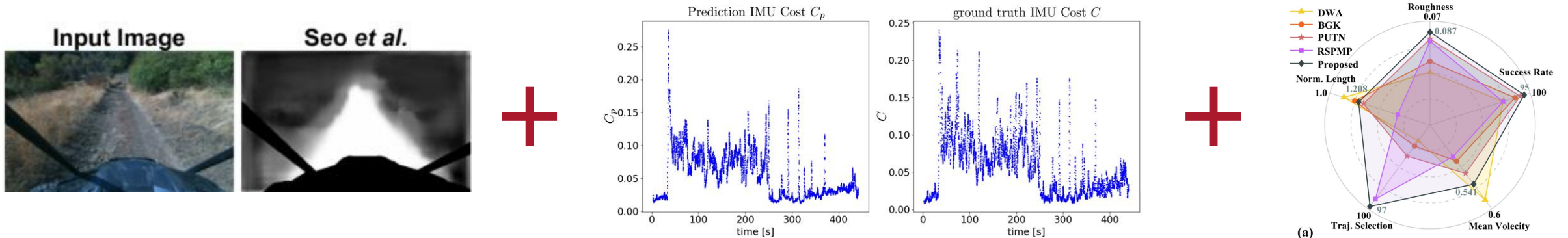
Problem Statement – Online Adaptation and Transfer

- Gather simulation data offline for both robots
 - Train models for both robots
- Gather online data for Scout Robot
 - Adapt Scout model
 - Transfer learning to Worker Robot model
 - Create traversability maps



Problem Statement – Evaluation

- Visual Inspection
- Baselines
 - No online adaptation
 - No transfer
 - Existing methods
- Model Error
- Objective Trajectory Metrics
 - Requires traversability optimizing motion planner



Questions?

